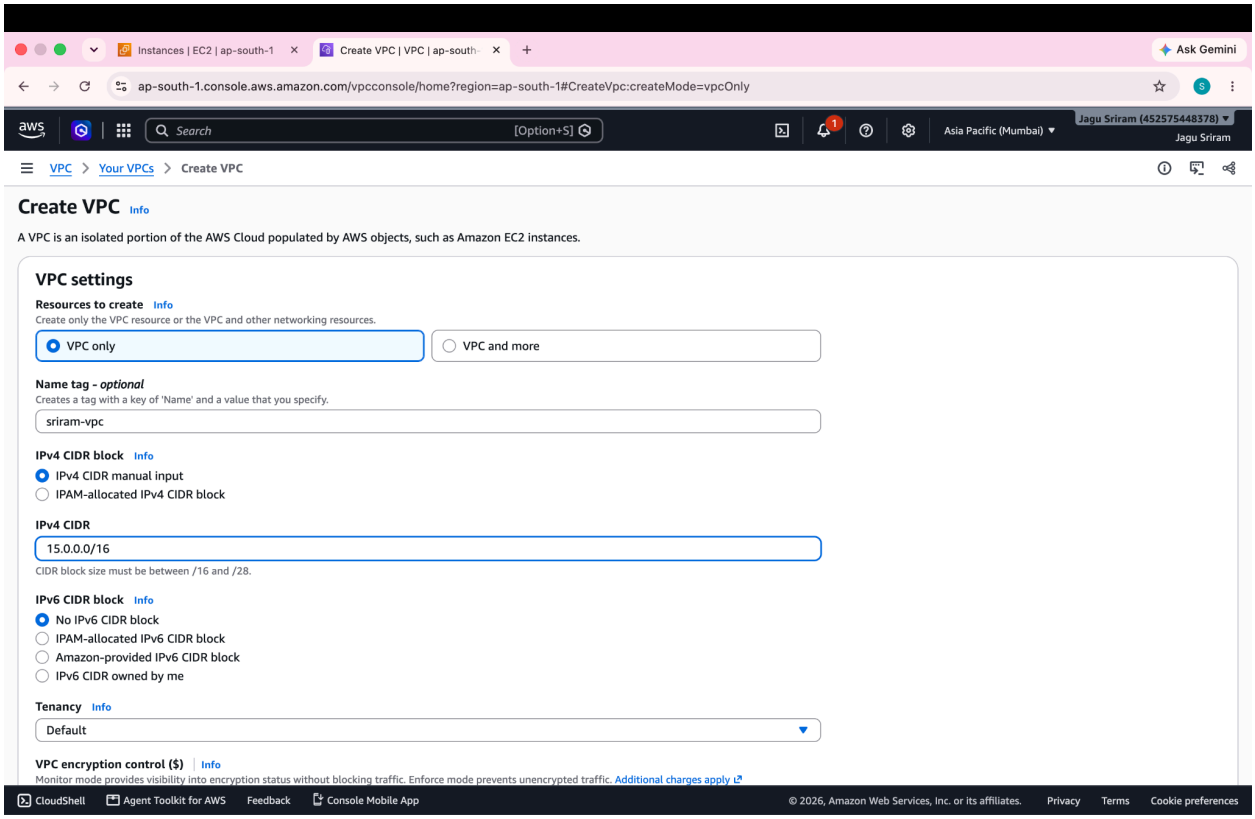
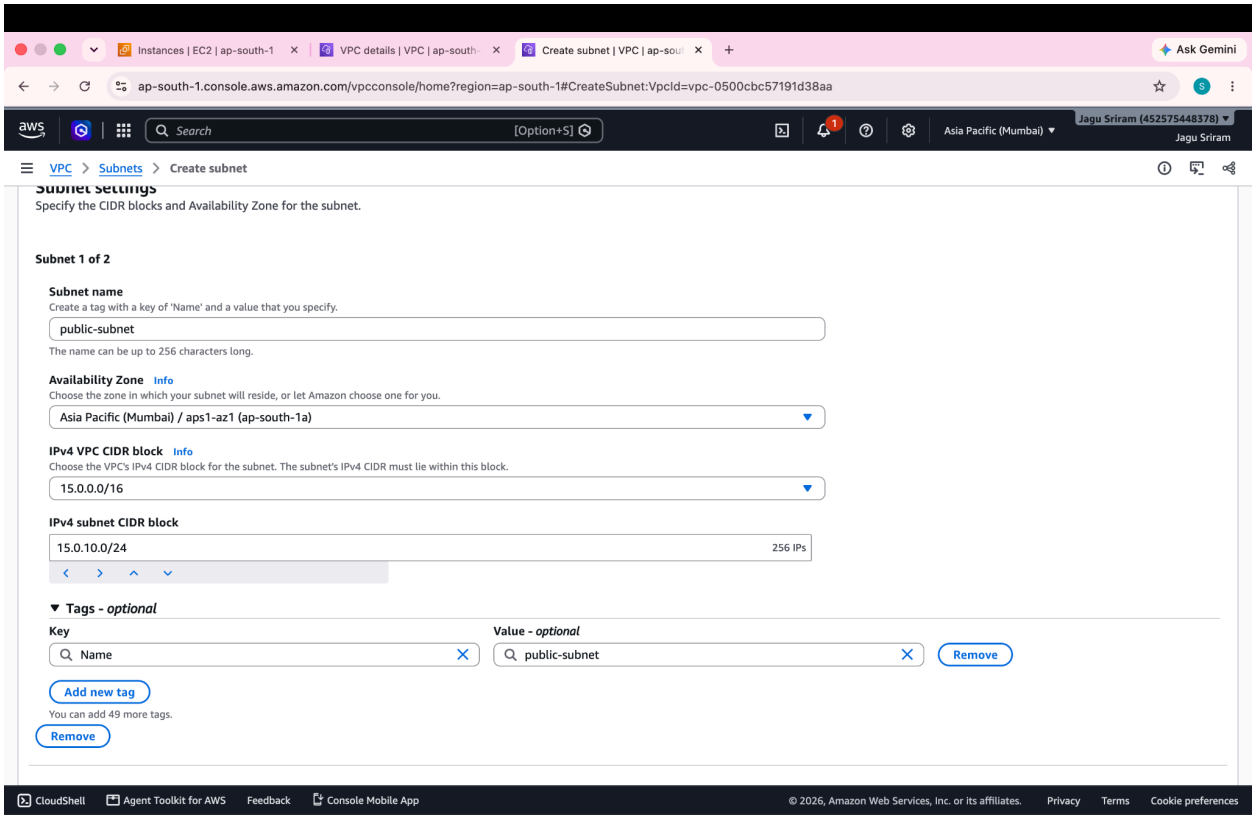
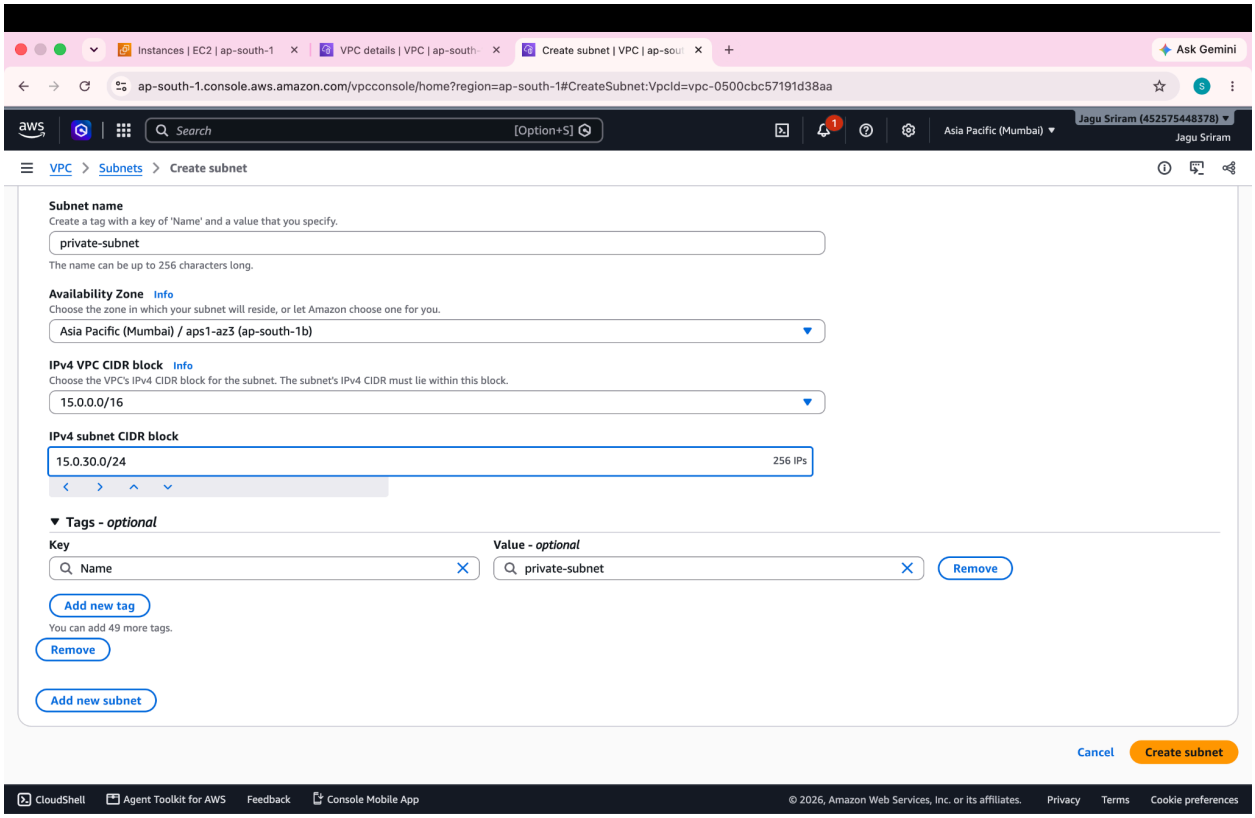
1. Creating VPC with CIDR block chosen as 15.0.0.0/16

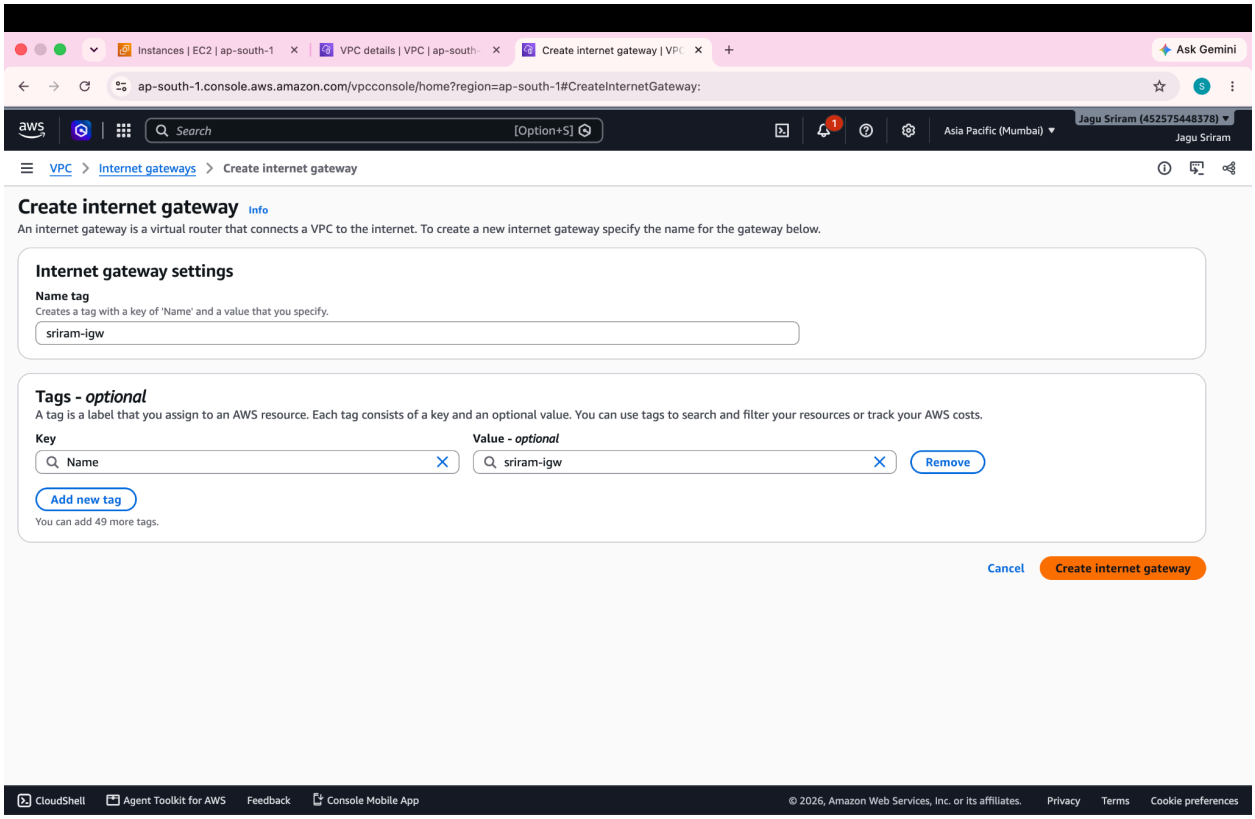


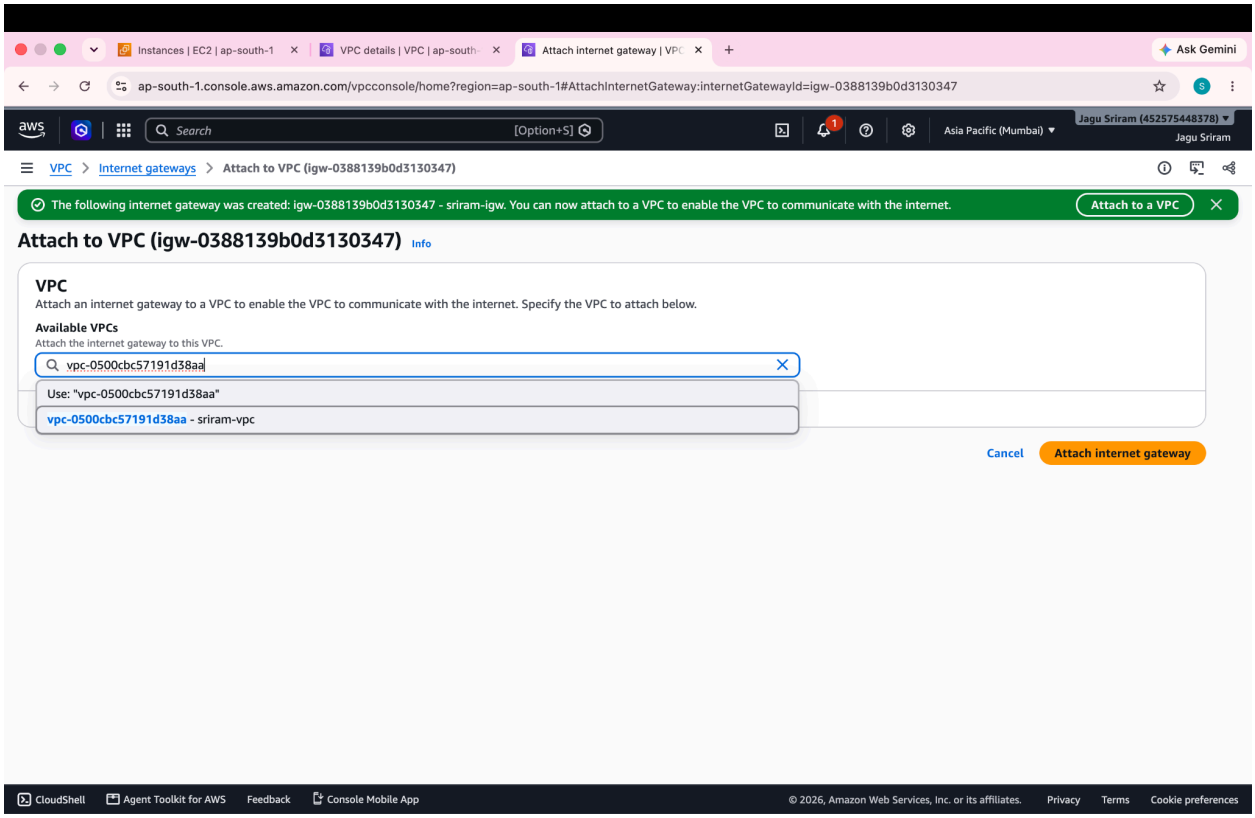
2. Creating 2 subnets (public-subnet and private-subnet):





3. Creating Internet Gateway and attaching it to VPC:

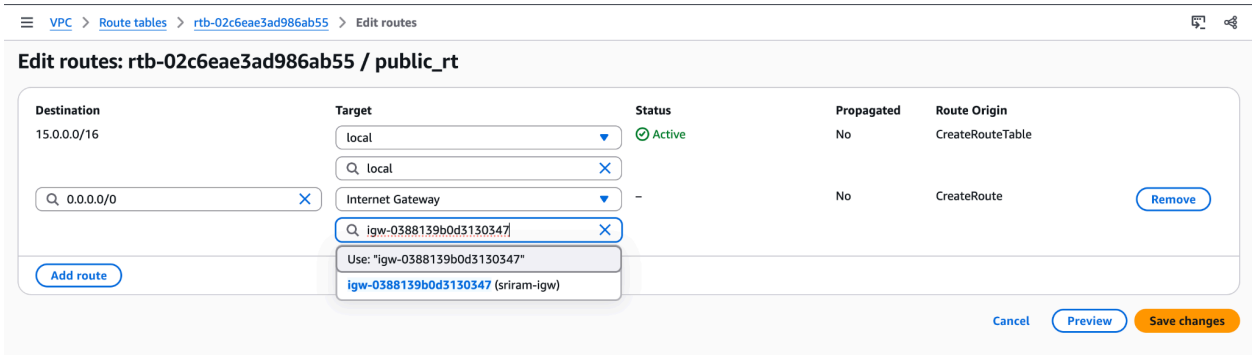




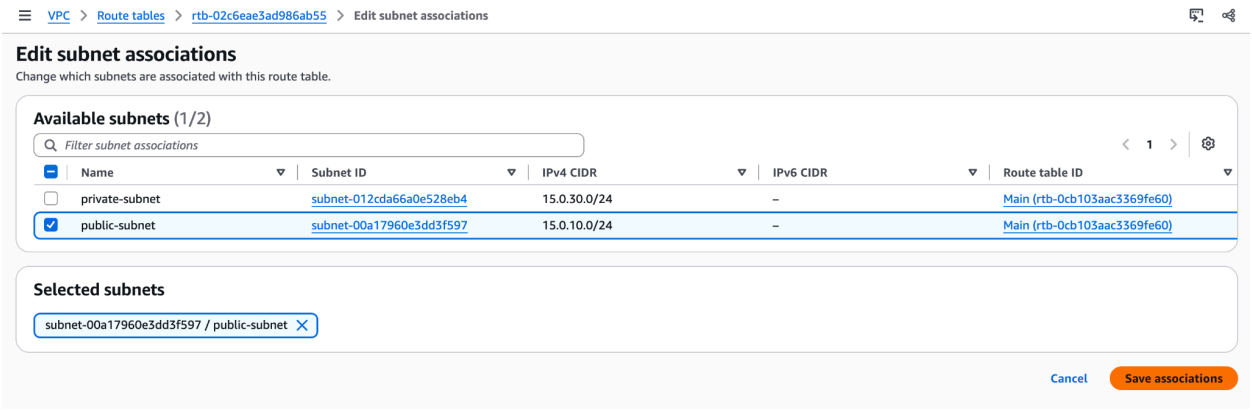
4. Created 2 route tables:



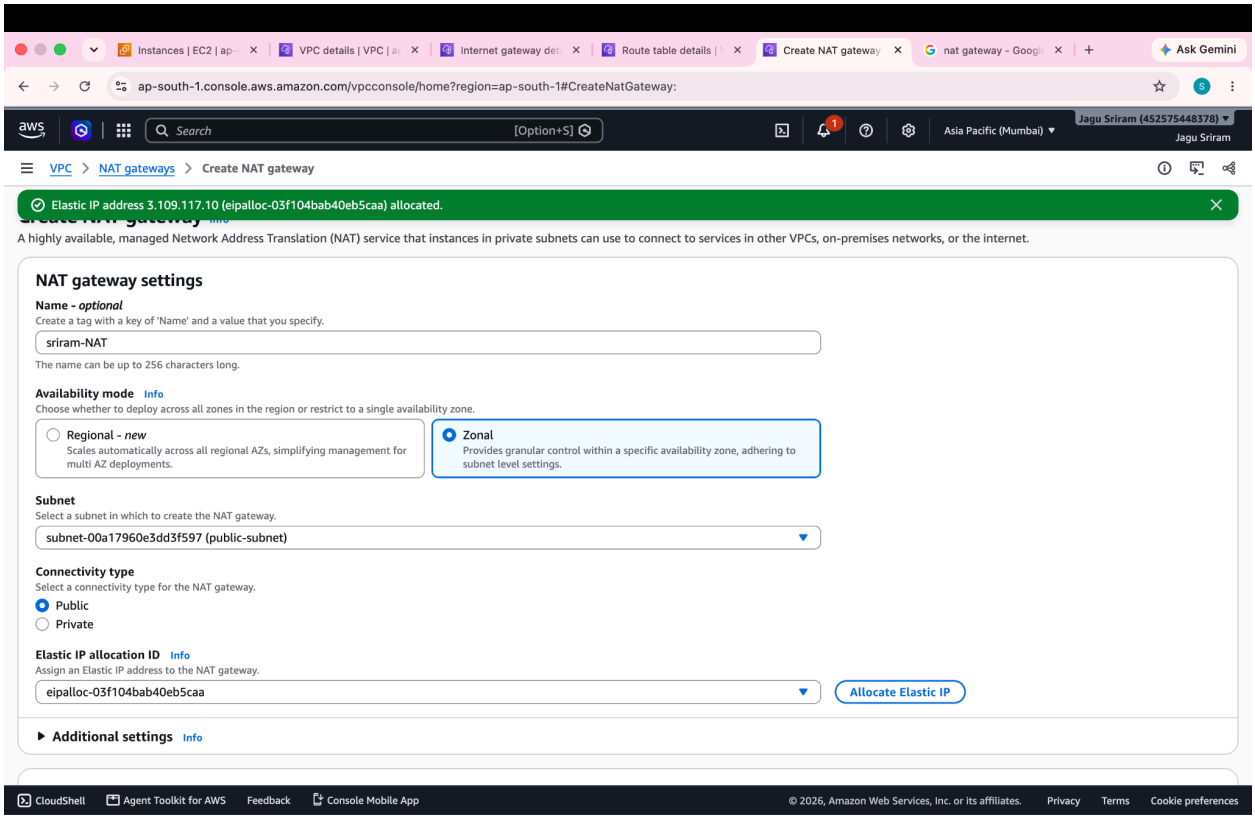
5. Adding internet Gateway route to a route table to make it public:



6. Associating public subnet to public route table:

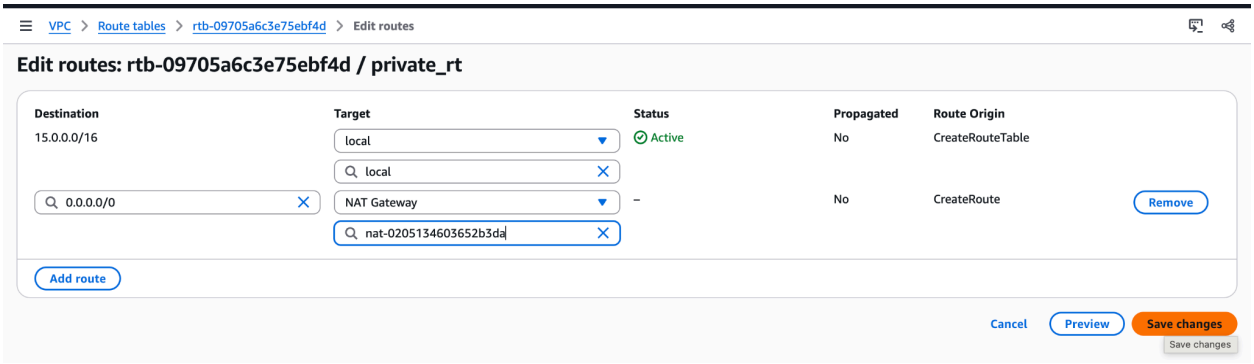


7. Creating NAT Gateway:
NAT Gateway allow resources in private subnet to access internet while blocking other unauthorized inbound traffic.
(Here we create NAT Gateway so as to associate with private subnet later)

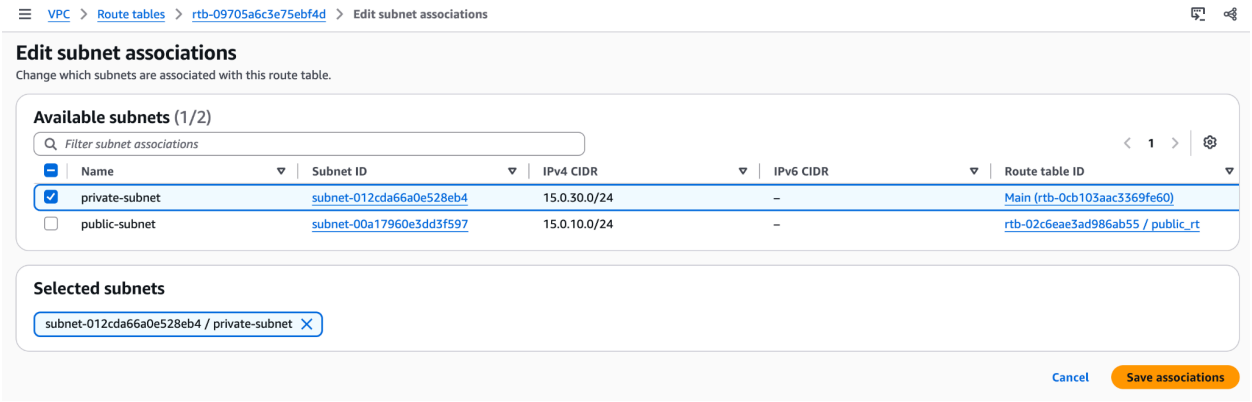


Here we attach a public IP address to a NAT gateway to enable private-subnet resources to communicate with the public internet while keeping them hidden from direct inbound access.
And here an elastic public IP is allocated because to provide static, predictable public IPv4 address for all outbound traffic coming from instances in private subnets.

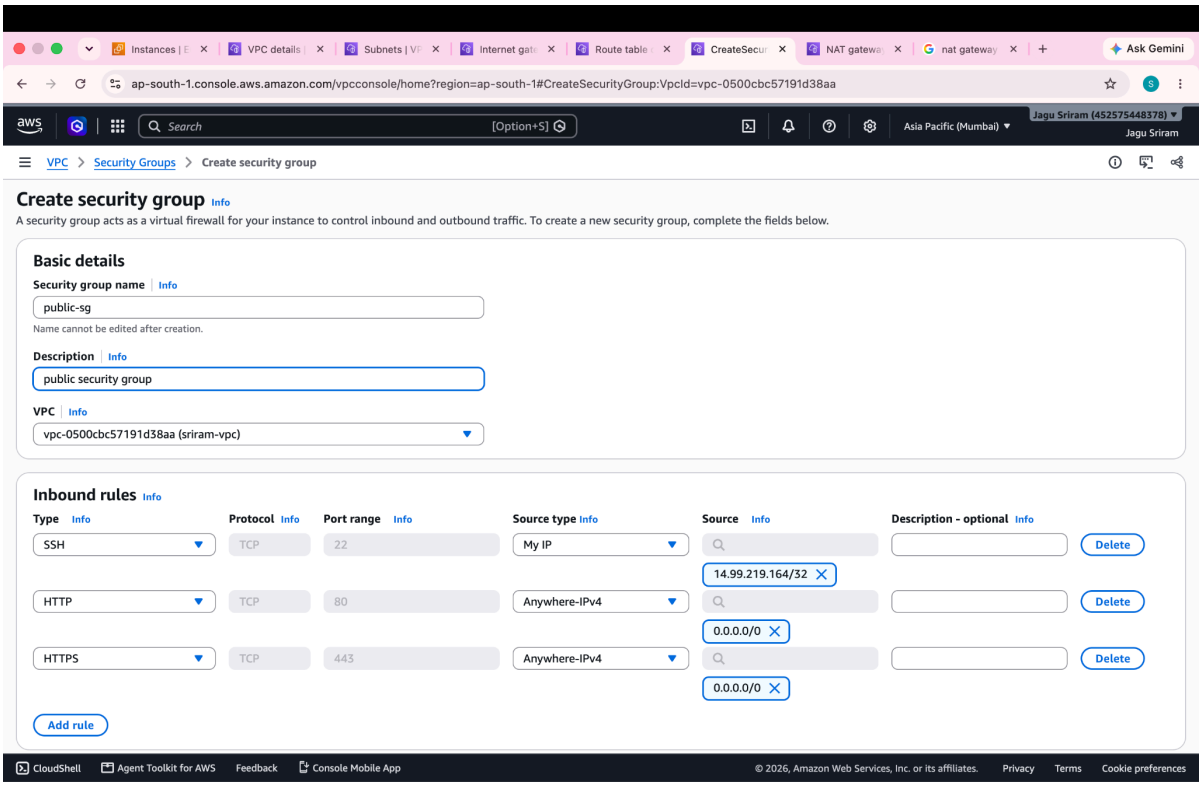
8. Now adding NAT gateway to private route table:



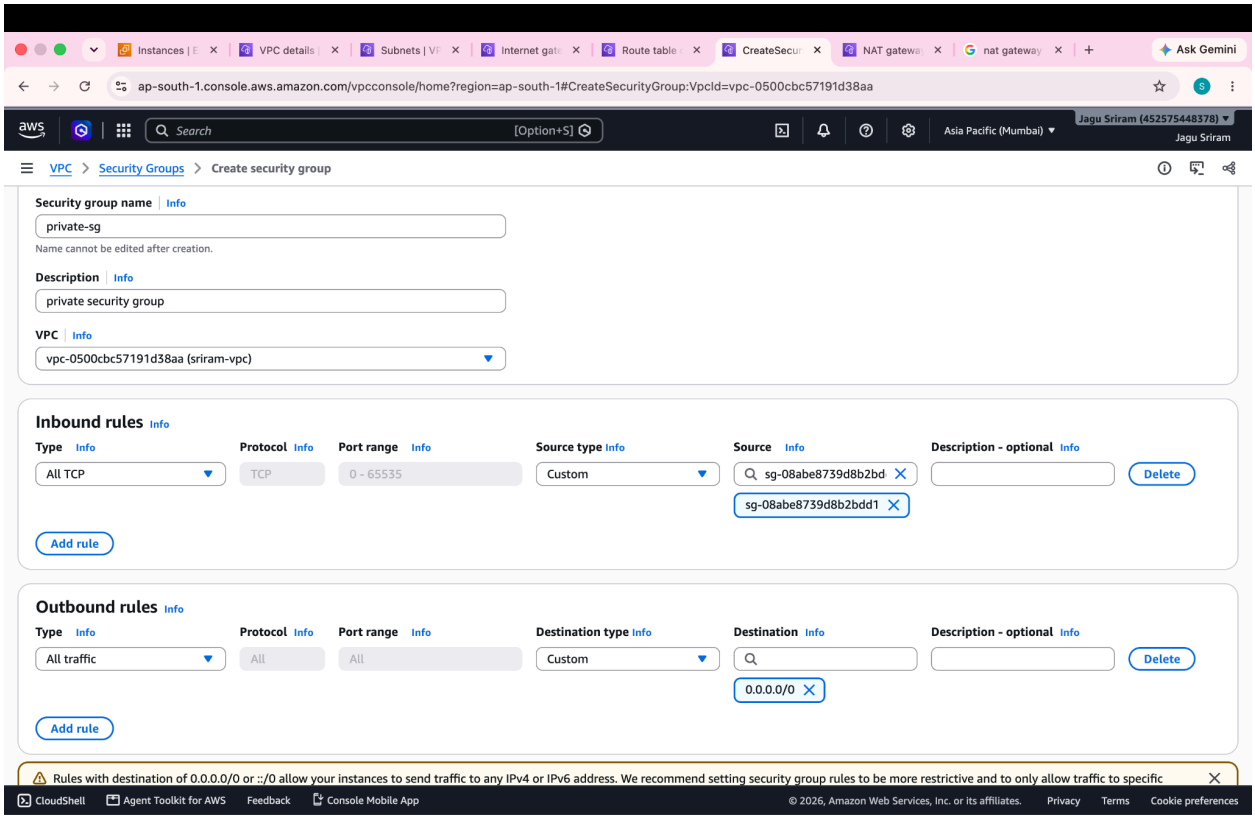
9. Associating private-subnet to private route table:



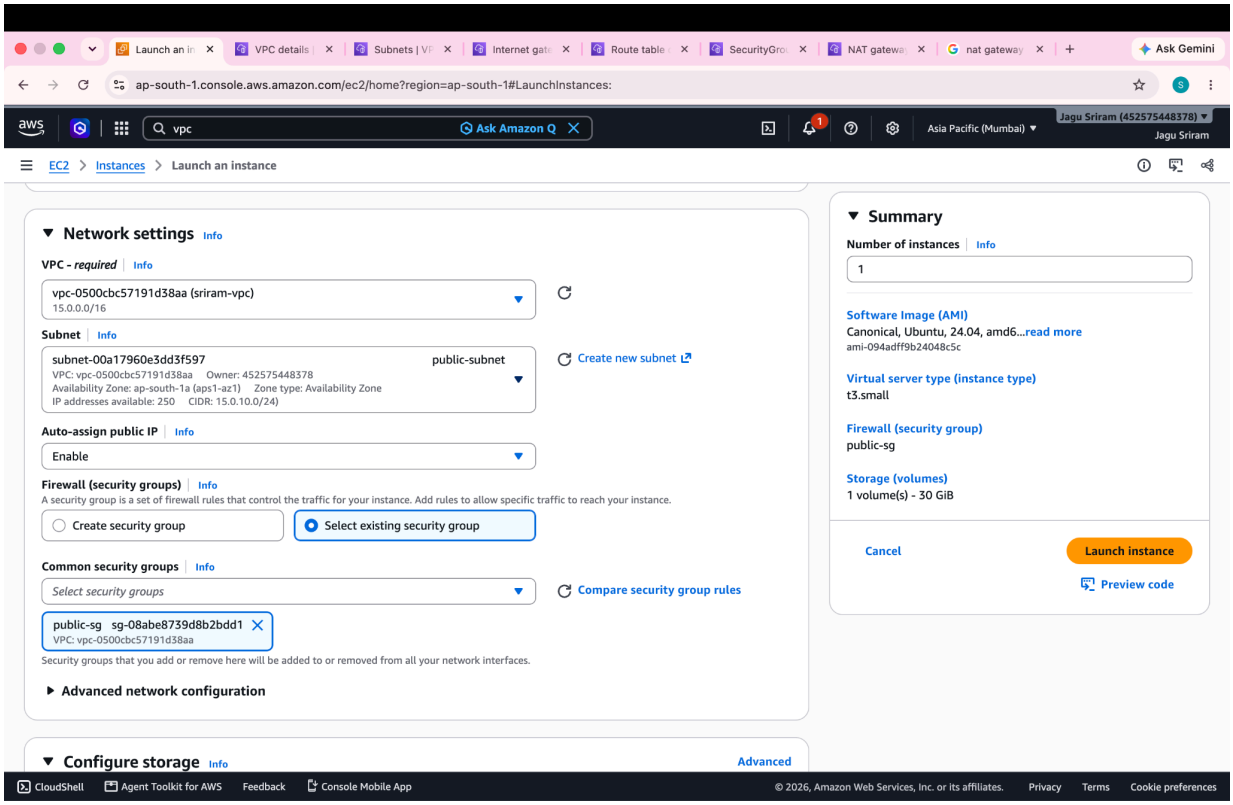
10. Creating public security group with below inbound rules set:



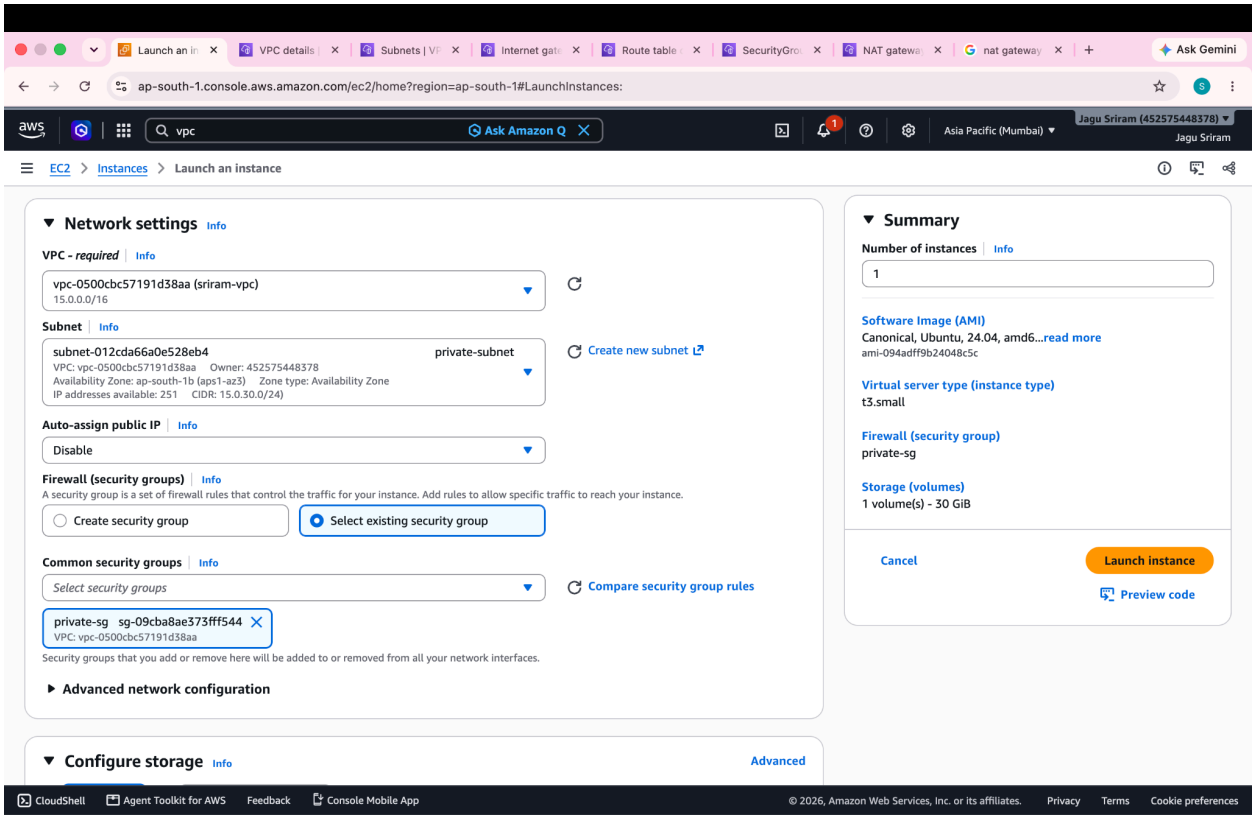
11. Creating private security group with inbound traffic set from public subnet:



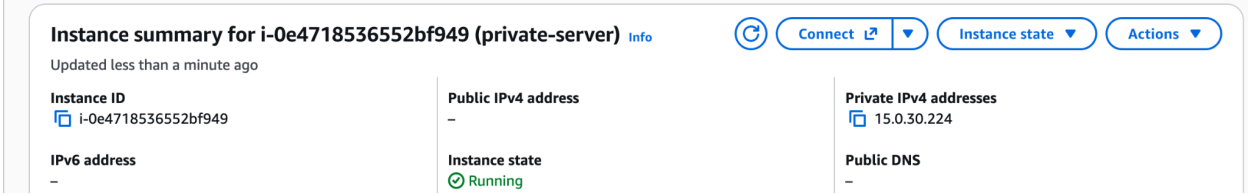
12. Creating public ec2 instance with network settings set as below:



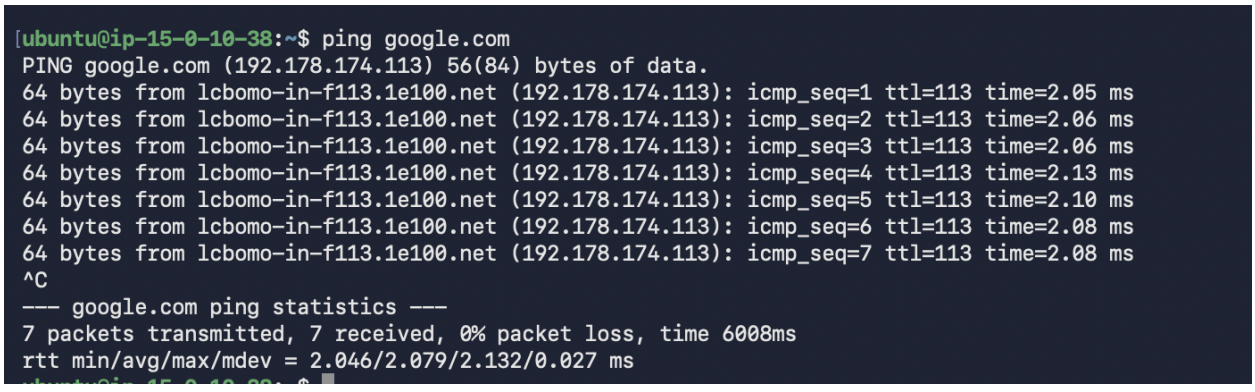
13. Creating private ec2 instance with network settings set as below:



Private server created without public IPv4 address:
Private IP: 15.0.30.224



14. Ping internet from public IP:



15. Adding pem file(private server pem file) to public server and set permissions and got access to private server:



16. Now private server can access internet via NAT gateway enabled through public server:
    The IP shown below is of private IP of private server.



17. Below screenshots prove that the private instance is accessing the internet through NAT as the public IP displayed matches the Elastic IP assigned to my NAT Gateway.

```
[ubuntu@ip-15-0-30-224:~$ curl https://checkip.amazonaws.com
3.109.117.10
```

nat-0205134603652b3da / sriram-NAT

Actions

Details

NAT gateway ID
nat-0205134603652b3da

NAT gateway ARN
arn:aws:ec2:ap-south-1:452575448378:natgateway/nat-0205134603652b3d

Connectivity type
Public

Primary public IPv4 address
3.109.117.10

State
Available

Primary private IPv4 address
15.0.10.84

State message Info
–

Primary network interface ID
eni-01525cbedc21ee663